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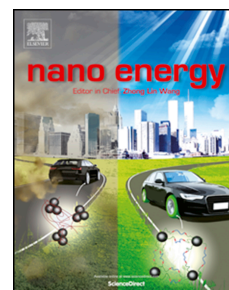
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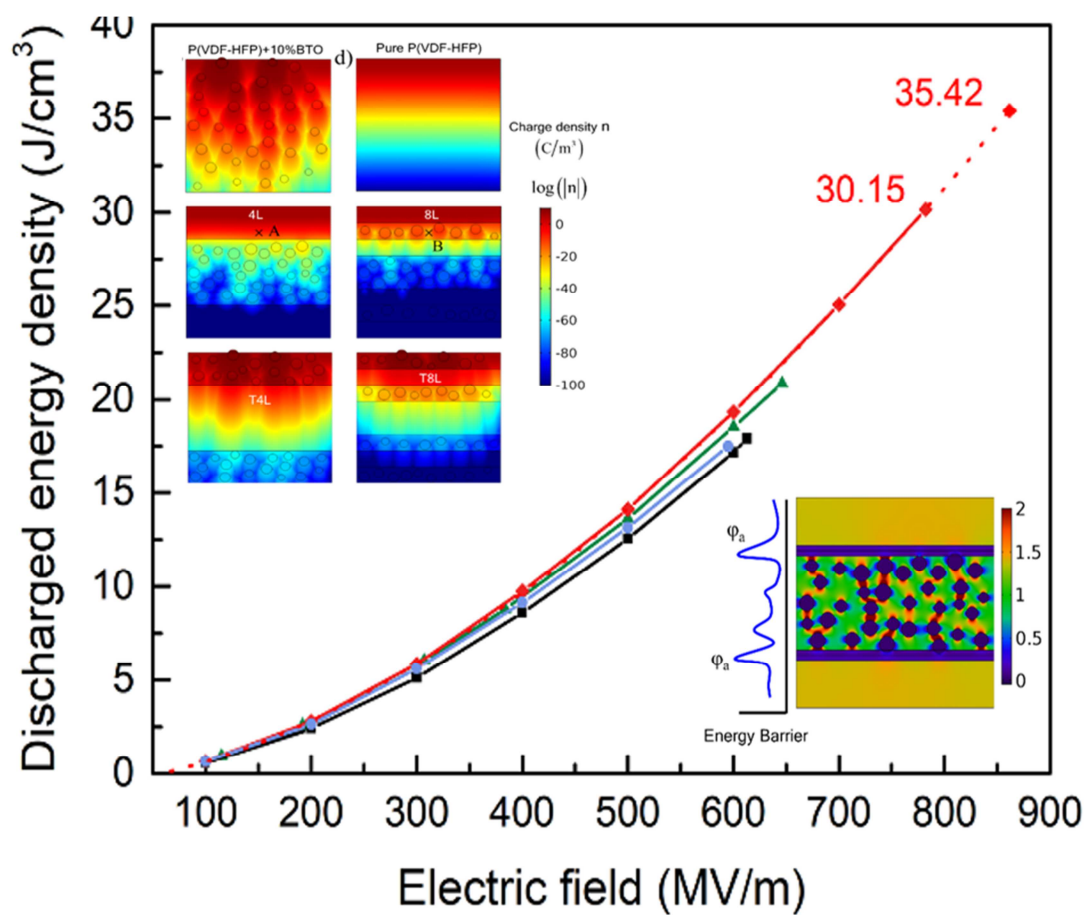
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Synergy of Micro-/Mesoscopic Interfaces in Multilayered Polymer Nanocomposites Induces Ultrahigh Energy Density for Capacitive Energy Storage

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Abstract: High-energy-density dielectric materials are highly desirable for the miniaturization and integration of modern electronics and power modules for applications in electrical power, communication, medical and defense systems. However, the conventional polymer nanocomposites with nanofillers randomly dispersed exhibit a limited energy storage performance (e.g. discharged energy density $< 15 \text{ J/cm}^3$ and efficiency $< 70\%$). Here, we demonstrate the multilayer structure as an effective route to polymer nanocomposites that

concurrently have ultrahigh discharge energy density and high efficiency. Compared to the random-dispersed nanocomposites, the rationally designed multilayered polymer nanocomposites are capable of integrating the suppressed effects of dielectric/electrode and dielectric/dielectric interfaces on charge injection and migration to remarkably enhance the breakdown strength and are expected to deliver an unprecedentedly high energy density of $\sim 35.4 \text{ J/cm}^3$ (an enhancement of $\sim 1100\%$ over the bench-mark biaxially-oriented polypropylene). This work provides insight into the design and fabrication of polymer nanocomposite with high energy density and discharge efficiency for capacitive energy storage applications.

KEYWORDS: PVDF, multilayer structure, energy density, interfaces, charge injection.

1. Introduction

Dielectric materials exhibit ultrahigh power density and are of great interest for a myriad of applications, notably in pulsed power systems, power conditioning and electrical weapon systems.[1-3] However, their relatively low energy density severely limits the continuing miniaturization of integrated circuits and the scaling-down of pulsed power systems.[4] In general, the discharged energy density (U_e) of dielectrics can be described by electric displacement (D) and electric field (E) as $U_e = \int_{D_{\max}}^0 E dD$. [5] Hence, high U_e will be achieved in dielectrics with high D (or high dielectric permittivity ϵ_r , due to $D = \epsilon_0 \epsilon_r E$, where ϵ_0 is the vacuum permittivity) and high breakdown strength E_b (the maximal E applied for a dielectric). Due to the adverse coupling between E_b and D ($E_b \sim D^{-0.5}$) in natural materials, both ceramics of high ϵ_r and

low E_b and pure polymer of high E_b and low ϵ_r fail to obtain a high U_e . [6] For instance, the bench-mark biaxially oriented polypropylene (BOPP) only exhibits a ϵ_r of 2.25, resulting in a low U_e of $\sim 5 \text{ J/cm}^3$ despite of their high E_b of $\sim 730 \text{ MV/m}$. [7]

In recent decades, polymer nanocomposites integrating the high ϵ_r of ceramic nanofillers and high E_b of polymer matrix have attracted significant attention and shown promises to achieve a high U_e , which is even comparable to that of the electrochemical capacitors. [1, 8, 9] So far, two types of polymer nanocomposites have been explored for their dielectric and energy storage behaviors, i.e. 0-3 type (with fillers randomly dispersed in polymer matrix) and 2-2 type (with multilayer structure). Due to the simple structure and easy preparation, the 0-3 type nanocomposites have been extensively investigated, and a host of nanofillers, such as 0D nanospheres (TiO_2 , BaTiO_3), 1D nanofibers/nanowires (SrTiO_3 , $\text{Ba}(\text{Zr}_x\text{Ti}_{1-x})\text{O}_3$) and 2D nanosheets (BN), etc. have been incorporated. [10-15] Indeed, a relatively high U_e can be achieved in some of these 0-3 type nanocomposites with cautious control. However, with the random dispersion of nanofillers, the dielectric breakdown paths could easily percolate throughout the nanocomposites especially with high loading of nanofillers, making it rather difficult if not impossible to further raise E_b of the 0-3 type nanocomposites. Compared to the 0-3 type nanocomposites, 2-2 type nanocomposites with multilayer structure are more effectively in modulating E_b and ϵ_r of nanocomposites in the following two aspects. [16-19] First, each layer in the multilayer-structured nanocomposites may function as either high E_b layer or high ϵ_r layer and can thus be independently regulated for the goal of rational design and concomitant

enhancement of E_b and ϵ_r . Second, in addition to microscopic interfaces between the filler and the polymer matrix which are universal for all polymer composites, another type of interface is present in the multilayer-structured nanocomposites on the mesoscopic scale, which is the interface between the adjacent dielectric layers (i.e. dielectric/dielectric interfaces). We proposed and demonstrated that these dielectric heterogeneous interfaces may induce substantially enhanced interfacial polarization and function as additional sites for charge storage.[16] Wang and co-workers also show that these dielectric/dielectric interfaces are also the region of weak electric field, which will hinder the growth of electrical trees and hence lead to significantly enhanced E_b . [17]

The other critical factors determining E_b and discharge efficiency of nanocomposites are the interfaces between the dielectrics and electrodes (i.e. dielectric/electrode interfaces). These interfaces have received much attentions recently because their structures and properties (e.g. the energy barrier for charge carriers) could have a great influence on the charge injection from electrodes and thus conduction behaviors and breakdown process in dielectrics. Li and Wang et al. have prepared sandwich-structured nanocomposites with layered c-BCB/BaTiO₃ (BT) and c-BCB/boron nitride nanosheets (BNNS). [18] The results showed that when the c-BCB/BNNS layer is close to electrodes, the energy barrier for charge injection is higher than that of c-BCB/BT case, leading to a lower conduction current and higher discharge energy density (U_e) and efficiency (η). Further, they recently demonstrated that a significantly improved high-temperature capacitive performance could be achieved in the current polymer films (e.g. BOPP, PEI, PC, etc.) sandwiched with SiO₂ layers. [20] For example, at 120 °C, U_e of

BOPP-SiO₂ ($\eta > 90\%$) is almost 18 times that of pristine BOPP (1.33 vs 0.076 J cm⁻³). It is attributed to the increase in the potential barrier at the dielectric/electrode interface induced by the wide-bandgap SiO₂ layer.

In sandwich-structured nanocomposites, the two dielectric/dielectric interfaces have a positive effect on dielectric performances. Then if the number of the dielectric/dielectric interfaces increases, the dielectrics (with multilayer structure) is reasonably believed to exhibit a better performance. Zhu and co-workers have investigated a number of pure polymer films with multilayer structure, such as PC/P(VDF-HFP), PSF/PVDF.[21-23] The results show that E_b of multilayered film is higher than that of each component or the blend counterparts.[21, 24] It is attributed to the interfaces that guide the electrical trees to propagate along the in-plane directions and thus extend the electrical trees. So far polymer nanocomposite films with multilayer structure have rarely been explored for their dielectric behavior at high electric field, especially for nanocomposites with more than 10 layers and a total thickness < 20 μm . The limitation mainly lies in two aspects. First, the melting co-extrusion method employed for the preparation of multilayered pure polymer films is not compatible with the preparation of composite films due to the severe agglomeration of nanofillers in the highly viscous polymer melt. Second, it is hard for the traditional solution-casting method to fabricate the multilayered nanocomposite films with the thickness of the individual layers < 2 μm . In this contribution, we demonstrate that by a nonequilibrium processing method that synergistically combines electrospinning, hot-pressing and thermal quenching, multilayered (up to 16 layers) polymer nanocomposites with interfaces of high structural integrity can be achieved. The multilayered

nanocomposites exhibit significantly improved dielectric performances, including greatly enhanced E_b , much suppressed conduction current and higher U_e . Besides, the effects of the number of layers and two types of interfaces on the microscopic and mesoscopic scale on the dielectric properties of multilayered nanocomposites are also investigated.

2. Results and discussion

2.1. Preparation of multilayered nanocomposites.

Figure 1a illustrates the nonequilibrium processing for the fabrication of the multilayered nanocomposites.[25, 26] This approach relies on the application of a modified electrospinning process to fabricate poly(vinylidene fluoride-hexafluoropropylene) (P(VDF-HFP)) nanofibers that are incorporated with BaTiO₃ (BTO) nanoparticles (~50 nm). As a result of the rapid evaporation of solvent during the electrospinning process, the BTO nanoparticles are “frozen” in the P(VDF-HFP) matrix in this nonequilibrium process hence are homogenously dispersed in P(VDF-HFP) nanofibers even without surface modification (**Figure 1b**; **Figure S1**, Supporting Information). Through a layer-by-layer process, P(VDF-HFP) pure nanofiber layer and P(VDF-HFP)-BTO composites nanofiber layer are stacked alternately into fibrous precursor mats (**Figure S1d**, Supporting Information), which are then transferred into dense nanocomposite films by a hot-pressing process. The obtained multilayered nanocomposite films are dense, smooth (**Figure 1c, d**) and have highly homogenous distribution of BTO nanoparticles (**Figure 1c, e**), indicating high structural integrity of the films. Here we prepare P(VDF-HFP)/P(VDF-HFP)-BTO multilayered nanocomposites with three different number of

layers, i.e. 4 layers, 8 layers and 16 layers, where the pure P(VDF-HFP) layer is close to the electrodes (the corresponding nanocomposites are denoted as 4L, 8L and 16L hereafter, **Figure 1f1-f3**). To explore the influence of the dielectric/electrode interfaces, three corresponding multilayered nanocomposites with reverse topological structure where the P(VDF-HFP)/BTO composite layer is close to the electrodes are also fabricated (denoted as T4L, T8L and T16L, **Figure 1g1-g3**). In addition, pure P(VDF-HFP) and 0-3 type P(VDF-HFP)-BTO nanocomposites with 5 vol.% and 10 vol.% BTO nanoparticles randomly dispersed in P(VDF-HFP) matrix are prepared as controls (**Figure S1&S2**, Supporting Information). Note that in a multilayered film, the thickness of individual layers is the same and the two same middle layers is designed to ensure same volume fractions (5 vol.%) for all multilayered nanocomposites (**Figure S3**, Supporting Information). As shown in **Figure 1c, 1f & 1g**, very clear, complete, flat and close-contact dielectric/dielectric interfaces are achieved in the multilayered nanocomposites even with 16 layers, signifying the ability of the nonequilibrium process to prepare nanocomposites with high structural integrity.

2.2. Dielectric performances of multilayered nanocomposites

Figure 2 summaries the dielectric properties and energy storage performances of multilayered nanocomposites. The frequency dependences of dielectric permittivity (ϵ_r) and dielectric loss ($\tan\delta$) of multilayered nanocomposites and the controls are presented in **Figure 2a**. As seen, the incorporation of high- ϵ_r BaTiO₃ nanoparticles yields an enhanced ϵ_r in multilayered nanocomposites with respect to pure polymer while $\tan\delta$ remains at low level. With increasing

number of stacking layers, multilayered nanocomposites exhibit mild increase in ε_r owing to the reduced influence of interfacial polarization at relatively high frequency, and meanwhile a continuing decreased dielectric loss.

The breakdown strength (E_b) of the nanocomposites is analyzed by the two-parameter Weibull statistics. The fitting function is described as $P(E) = 1 - \exp[-(E/E_b)^\beta]$, where $P(E)$ is the cumulative probability of electric failure, E is the experimental breakdown strength, the scale parameter E_b is the characteristic breakdown strength corresponding to a 63.2% probability of failure and the shape parameter β is the Weibull modulus that evaluates the dispersion of E . [27] The fitting results are presented in **Figure S4** (Supporting Information) and the characteristic breakdown strength are summarized in **Figure 2b**. As seen, compared with the pure P(VDF-HFP) film, E_b of 0-3 type nanocomposites drops continuously with increasing amount of BTO nanoparticles. However, the nanocomposites with 5 vol.% of BTO nanoparticles still deliver a E_b of ~ 595.1 MV/m (much higher than that of traditional solution-casting nanocomposite films), demonstrating high quality of the films and homogenous dispersion of BTO nanoparticles. [28, 29] By constructing the nanocomposites with 4 layers (actually sandwich structure), E_b of the 4L films obviously increases to ~ 646.6 MV/m compared to the 0-3 type nanocomposites with the same 5 vol.% of BTO nanoparticles loading, while a decrease in E_b is observed in the T4L films (~ 548.6 MV/m) with reverse topological structure. With increasing number of layers, there is a prominent enhancement of E_b in both multilayered nanocomposites and their counterparts with reverse topological structure. For instance, E_b of the 16L films reaches up to ~ 862.5 MV/m, which represents a $\sim 33.3\%$ and $\sim 40.5\%$ enhancement over those of

the 4L films and the pure P(VDF-HFP) films (~ 613.7 MV/m). To our best knowledge, this is so far the highest E_b ever achieved for any polymer nanocomposites filled with high- ϵ_r nanoinclusions. Also worth noting is that all of nanocomposites exhibit rather high Weibull modulus β of ~ 20 -30 (**Figure S4**, Supporting Information), which is even comparable to the pure polymer thin films fabricated with a extrusion-stretching process and indicates high dielectric reliability of these multilayered nanocomposites.

In general, the dielectric permittivity is measured to evaluate the weak-field dielectric polarization of the dielectrics, and their electric polarization and energy storage performances should be characterized by their electric displacement – electric field (D - E) loops at high electric field. **Figure 2c** compares the maximum electric displacement (D_{max}) derived from the D - E loops (**Figure S5**, Supporting Information) of the nanocomposites with different nanostructures (at E_b , except for the 16L sample). As seen, D_{max} of 0-3 type nanocomposites increases gradually with increasing BTO loading, which could be attributed to the high dielectric permittivity of the BTO nanoparticles. For the multilayered nanocomposites with the same feeding ratio of 5 vol.%, a pronounced enhancement in D_{max} is achieved in comparison with their 0-3 counterparts, where D_{max} increases sharply with increasing number of stacking layers. For instance, D_{max} for the 4L films is $\sim 9.09 \mu\text{C}/\text{cm}^2$, compared to $\sim 8.47 \mu\text{C}/\text{cm}^2$ for the 0-3 type nanocomposites with 5 vol.% of BTO nanoparticles, and reaches up to $\sim 10.23 \mu\text{C}/\text{cm}^2$ for the 8L films and $\sim 11.01 \mu\text{C}/\text{cm}^2$ for the 16L films, respectively. It is worth noting that D_{max} of 16L is obtained at ~ 782 MV/m, which is the upper limit of our current equipment for D - E loop measurements but still is lower than its E_b . We therefore anticipate that even higher D_{max} could be obtained at E_b of the 16L films.

However, despite that, this value is still higher than those of P(VDF-HFP)/BTO@TO_nfs nanocomposites and P(VDF-HFP)/BTO nanocomposites with fiber-orthotropic structure reported in our previous work (Figure 2c).[9, 26] The much enhanced high-field electric polarization in the multilayered nanocomposites could first be attributed to the markedly enhanced breakdown strength, which allows the multilayered nanocomposites to be polarized at higher electric field according to $D = \epsilon_0 \epsilon_r E$. Secondly, it could be induced by the substantially enhanced interfacial polarization at the dielectric/dielectric interfaces, as evidenced by the enhanced D_{max} in the 16L films compared to the 4L films at the same electric field (**Figure S6**, Supporting Information). For example, at 500 MV/m, D_{max} rises from $\sim 7.8 \mu\text{C}/\text{cm}^2$ for the 4L films to $\sim 8.3 \mu\text{C}/\text{cm}^2$ for the 16L films.

According to $U_e = \int_{D_{max}}^0 E dD$, concomitantly enhanced E_b and D_{max} in multilayered nanocomposites give rise to a higher discharged energy density (U_e). **Figure 2d** and **Figure S7** (Supporting Information) exhibit the U_e and charge-discharge efficiency (defined as $\eta = U_e/U$, where U is the total electric energy density stored during the charging process of dielectrics) of all nanocomposites. As seen, the 16L films deliver a rather high U_e of $\sim 30.15 \text{ J}/\text{cm}^3$ at a field of $\sim 782 \text{ MV}/\text{m}$, which represents a $\sim 145\%$ improvements over the 4L film ($\sim 20.85 \text{ J}/\text{cm}^3$) and $\sim 168\%$ over pure P(VDF-HFP) ($\sim 17.90 \text{ J}/\text{cm}^3$) at E_b . Note that the maximal U_e of the 16L films is measured below E_b , which is limited by the upper limit of our equipment. With rational fitting the $U_e - E$ curve, an unprecedented high U_e of $\sim 35.42 \text{ J}/\text{cm}^3$ is expected at E_b in the 16L films, which is by far the highest U_e for polymer nanocomposites. Constructing multilayered structure is also an effective strategy for the reverse configuration, as evidenced by the enhanced U_e

shown in **Figure S7c** (Supporting Information). The charge-discharge efficiency is another critical figure of merit of dielectric materials for actual applications, especially at high electric field. For most dielectric films of current use, η drops gradually with increasing electric field to a relative low level at high electric field. For instance, η of melt-stretched P(VDF-HFP) films is only ~57% at 300 MV/m.[30] In a stark contrast, the 16L film shows a much higher η of ~78% even at very high electric field of ~800 MV/m, demonstrating the marked energy storage performances of the multilayered nanocomposites. To further highlight the significance of our multilayered strategy, **Table 1** shows the comparison of discharge energy density and efficiency in this work and the related polymer nanocomposites reported in the previous literatures.[2, 12, 17, 28, 31-33] One can see that the multilayered nanocomposites exhibit significantly higher discharge energy density and efficiency compared with other listed polymer nanocomposites, demonstrating the superiority of the multilayered structure in enhancing the energy storage performance for polymer dielectrics.

2.3. Thermally stimulated discharge current (TSDC) and phase-field simulations

The impressively improved energy storage performances in multilayered nanocomposites are mainly ascribed to the greatly enhanced breakdown strength and the much enhanced electric polarization. Besides, the energy storage performances of the multilayered nanocomposites are superior to that of their counterparts with reverse topological structure. Given their different configurations, we believe that the prominent difference in energy storage behaviors could be related to their differences in the two kinds of interfaces, e.g., dielectric/dielectric and

dielectric/electrode interfaces. In an effort to understand the relationship between the enhanced performances and the properties of interfaces, multilayered nanocomposites and the controls are characterized by TSDC technology and simulated by a phase-field model.

TSDC spectroscopy is employed to detect charge injection from electrodes and interfacial polarization of space charges in the multilayered nanocomposites.[34, 35] **Figure 3a** shows the TSDC spectroscopies of the multilayered nanocomposites and the controls, which are collected under a poling field (E_p) of 15 MV/m for a poling time of 30 min at poling temperature (T_p) of 50 °C. After peak deconvoluting, four discharge peaks could be identified at -42 °C (peak 1), -15 °C (peak 2), 58 °C (peak 3) and 75-95 °C (peak 4), respectively. Peak 1 is assigned to the depolarization of the polarized amorphous P(VDF-HFP) via devitrification at T_g (glass transition temperature), and peak 2 could be related to molecular motions in the crystalline region.[27, 34] We consider that peak 3 at ~58 °C should be ascribed to the depolarization of the injected charges from electrodes, i.e., Schottky or thermionic emission.[36] To verify this hypothesis, the E_p -dependent TSDC test is performed for pure P(VDF-HFP) (**Figure S8a**, Supporting Information). In general, for Schottky emission, the current density (J) scales as $\ln J \propto E_p^{1/2}$. [36] At equilibrium state, the injected charges equal to the leakage charges (C), which can be obtained from the TSDC spectrum as $\ln C \propto E_p^{1/2}$. [34] **Figure S8b** (Supporting Information) shows the linear relationship between $\ln C$ and $E_p^{1/2}$, which supports our prediction. The discharged peak 4 could be attributed to the depolarization of ions, because it appears at relatively high T_p (**Figure S9**, Supporting Information).[35] The ions with smaller mobility compared to the electrons could be too slow to be polarized when the poling temperature is low.

Here, we pay more attention to the information obtained from peak 3 and 4. The areas and center temperatures of peak 3 and 4 are summarized in **Figure 3b**. As seen, for 0-3 type nanocomposites, the area of peak 3 increases with increasing feeding ratio, indicating enhanced charge injection after incorporating high-permittivity nanofillers. Accordingly, the area of peak 3 for the multilayered nanocomposites where the pure P(VDF-HFP) layer is close to the electrodes is smaller than that of their counterparts with the composite layers filled with 10 vol.% of BTO nanoparticles close to the electrodes. Plus, charge injection from the electrodes is suppressed with increasing number of stacking layers in both of the two configurations, as evidenced by the decreased area of peak 3. As for peak 4, two striking features could be distinguished. First, the area of peak 4 for the multilayered nanocomposites is consistently higher than that for the 0-3 type nanocomposites with the same feeding ratio of 5 vol.%, suggesting that more ions are captured into the traps during the poling process and then could be thermally stimulated to form a larger discharged current. Second, the area of peak 4 increases with increasing number of stacking layers. We are thus led to the conclusion that the ions in the 0-3 type nanocomposites are distributed at either the nanofiller/polymer interfaces or amorphous/crystal interfaces, and the extra ions in the multilayered nanocomposites may originate from the traps at the dielectric/dielectric interfaces that are present only in the multilayered nanocomposites. Therefore, they will increase with the area of interfaces. Considering the quite small part of ions distributed at other places (i.e. the nanofiller/polymer interfaces or amorphous/crystal interfaces in the polymer matrix away from dielectric/dielectric interfaces), peak 4 is mainly attributed to the depolarization of dielectric/dielectric interfacial ions.[35] Note that the multilayered

nanocomposites exhibit higher peak 4 compared to their counterparts with reverse topological structure. For example, the area of peak 4 in the 4L film is $\sim 0.42 \mu\text{C}$, while it is only $\sim 0.12 \mu\text{C}$ for the T4L film. We speculate that it may result from the recombination of interfacial ions with injected charges during the poling process. That is, more charges are injected from the electrodes (like in the T4L case), more interfacial ions will be recombined during the poling process and hence smaller discharge current (smaller area of peak 4) will be obtained during the following thermally stimulating process. It is also one of the reasons for the larger area of peak 4 achieved in the multilayered nanocomposites with more layers that have a weaker charge injection. Also of interest to note is that the center of peak 4 shifts to higher temperature as the number of stacking layers increases, which suggests that the depth of the traps at the dielectric/dielectric interface also increases. The ions trapped at the interfaces establish a strong depolarization electric field (as illustrated in **Figure 4a**) that is opposite to the external electric field, which could hamper the motion of charge carriers along the direction of the external electric field. Combined with the much suppressed charge injection from the electrode, the slower motion of charge carriers leads to remarkably decreased leakage current in the 16L films as shown in the results of leakage current tests (**Figure 3c**). Due to more pronounced charge injection from the electrodes, the multilayered nanocomposites with reverse topological structure exhibit higher leakage current (**Figure S10**, Supporting Information). For example, at 150 MV/m, the leakage current is $\sim 1.42 \times 10^{-6} \text{ A/cm}^2$ for the 4L films and $\sim 3.43 \times 10^{-6} \text{ A/cm}^2$ for the T4L films.

To further understand the charge injection and mobility behaviors of charge carriers in nanocomposites, we employ a phase-field model to qualitatively simulate the charge distribution

in nanocomposites with various configurations with consideration of interfacial polarized ions (**Figure 4**).[37] As seen, when relatively high volume fraction of BTO nanoparticles are randomly dispersed in P(VDF-HFP), much higher charge density appears in **Figure 4c** compared with pure P(VDF-HFP) (**Figure 4d**), demonstrating a much severe charge injection. Since BTO nanoparticles are less insulating and of higher electron mobility than polymer matrix, high loading of BTO nanoparticles will form a faster path for charge carriers to migrate in dielectrics and also promote charge injection from electrodes. Comparison of the simulation results of the 4L (**Figure 4g**) and T4L (**Figure 4j**) nanocomposites show much higher charge density appear in the T4L sample as a result of much severe charge injection, which is in line with the larger area of peak 3 in the TSDC spectra (**Figure 3b**). Due to the much severe charge injection, the multilayered nanocomposites with reverse topological structure exhibit higher leakage current, lower E_b and U_e . As the number of stacking layers increases, the charge injection is weaker when comparing the charge distribution of the 4L films (**Figure 4g**) and the 8L films (**Figure 4h**), which also agrees with the TSDC results. For example, the charge density of point A (**Figure 4g**) in the 4L films is $\sim 10^{-15}$ C/cm³, which is orders of magnitudes higher than that of point B (**Figure 4h**) at the same position in the 8L films ($\sim 10^{-20}$ C/cm³). In the multilayered nanocomposites, the polarized ions at the dielectric/dielectric interfaces will induce a strong depolarized field (E_d) opposite to the applied electric E^{app} (**Figure 4a, f, i**). The E_d will lower the local electric field E^{local} at the interfaces as shown in **Figure 4e**, and increase the height of potential barrier ϕ_a for the migration of charge carriers across the interfaces. Thus, the charge migration in the 4L films becomes harder and the charge injection is suppressed, as evidenced by

the lower charge density in **Figure 4g** compared to the pure P(VDF-HFP) (**Figure 4d**). In the T4L films, although the P(VDF-HFP)/BTO composite layer close to the electrodes causes a high charge density near the electrodes, the E_d is also beneficial to decrease the charge density and alleviate the charge injection by comparing **Figure 4c** and **Figure 4j**. In general, the suppressed number and motion of charge carriers will not only lead to a reduced leakage current (**Figure 3c**), but impede the electrical treeing process, making the dielectrics less susceptible to electrical breakdown. It is also noted that the Young's modulus of the multilayered nanocomposites measured by nanoindentation exhibits marginal change with varied number of stacking layers (**Figure S11**, Supporting Information). It is therefore reasonable to infer that electrical breakdown rather than electromechanical breakdown plays a more critical role in breakdown process for the multilayered nanocomposites. Therefore, the enhanced hindrance to the electrical breakdown may be the source of the greatly improved E_b and U_e for the multilayered nanocomposites with more layers (within 16 layers).

3. Conclusions

In summary, multilayered polymer nanocomposites with high quality and high integrity are fabricated by the nonequilibrium processing method. Through tuning the number and order of the stacking layers in the multilayered nanocomposites, the effects of dielectric/electrode and dielectric/dielectric interfaces on the dielectric performances of the multilayered nanocomposites are investigated by TSDC spectra and phase-field simulation. The results suggest that highly

insulating pure P(VDF-HFP) are more effective in suppressing the charge injection from electrodes when close to the electrodes as compared with the P(VDF-HFP)/BTO composite layer. Furthermore, the polarized ions located in dielectric/dielectric interfaces will generate a strong depolarization electric field and act as barrier to suppress the motion of charge carriers across the interfaces hence hinder the propagation of electrical trees, which yields the greatly improved E_b , higher D_{max} , and hence drastically enhanced U_e . In comparison to the 0-3 type nanocomposites, the multilayered nanocomposites with two types of interfaces deliver more superior energy storage performances. This work also presents new insights into the mechanisms about charge transportation in multilayered nanocomposites, and offers a new design paradigm of high-energy-density polymer nanocomposites.

4. Experimental Section

4.1. Materials

P(VDF-HFP) powders (Arkema, Kynar Flex 2801, with 10 wt.% of HFP) were used as received. BaTiO₃ nanoparticles (~50 nm in diameter) were purchased from Sinocera Corporation Ltd. All solvents were obtained from China National Chemicals Corporation Ltd. and used without further purification.

4.2. Fabrication of P(VDF-HFP)/P(VDF-HFP)-BTO multilayered nanocomposites

The non-equilibrium process, combining the electrospinning, hot-pressing and thermal treatment, was employed to fabricate the multilayered nanocomposites. Specifically, P(VDF-HFP) powders

were thoroughly dissolved in the mixed solvent of *N,N*-dimethylformamide (DMF) and acetone to obtain pure P(VDF-HFP) precursor solution. With the aid of ultrasonication and magnetic stirring, BaTiO₃ nanoparticles were homogenously dispersed in P(VDF-HFP) solution to get P(VDF-HFP)-BTO precursor solution (with 10 vol.% BTO loading). Two preformed precursor solutions were then transferred into two syringes, respectively. The multilayered fibrous mats were fabricated through a modified electrospinning. Take the preparation of the 4L sample for example, the first layer, pure P(VDF-HFP) fibers, were electrospun using pure P(VDF-HFP) precursor with an electric field of 1 kV/cm and flow rate of 1 mL/h. Then, second to fourth layers, i.e., P(VDF-HFP)-BTO composite fiber layer, P(VDF-HFP)-BTO composite fiber layer and P(VDF-HFP) fiber layer were electrospun with same electrospinning conditions. The similar process was applied to fabricate other multilayered fibrous mats (8L, 16L, T4L, T8 and T16L), except for different stacking orders and electrospinning time for individual layer. The as-spun fibrous mats were hot-pressed at 200 °C for 30 min under a pressure of 10 MPa and then the as-pressed films were annealed at 200 °C for 15 min followed by quenching in ice water immediately. All of the multilayered films had equal volume fraction of BTO nanoparticles (fixed at 5 vol.%) and an overall thickness of 13 μm.

4.3. Fabrication of P(VDF-HFP) and P(VDF-HFP)-BTO nanocomposites

The pure P(VDF-HFP), P(VDF-HFP)-5 vol.% BTO and P(VDF-HFP)-10 vol.% BTO fibrous mats were electrospun using the corresponding precursor solutions under the same electrospinning condition described as above, respectively. Then, the same hot-pressing and thermal treatment were applied to obtain the final films as controls.

4.4. Characterization

The morphology of P(VDF-HFP) nanofibers, P(VDF-HFP)-BTO nanofibers and multilayered nanocomposite films were characterized with scanning electron microscopy (SEM, ZEISS MWRLIN compact). Differential scanning calorimetry (DSC, TA Q-2000) was performed for all samples. Dielectric permittivity and dielectric loss were tested by a precision impedance analyzer (Agilent Technologies, HP 4294A) at 1 V_{rms} and room temperature within the frequency range of 10² to 10⁷ Hz. Electric breakdown tests were performed with dielectric withstand voltage test system (Beijing Electro-mechanical Research Institute Supesvoltage Technique; 20 kV) at a ramping rate of 200 V/s and a limit current of 5 mA. Unipolar displacement-electric field (*D-E*) hysteresis loops were measured at 10 Hz with a Premier II ferroelectric test system (Radiant Technologies, Inc.; 10 kV). The leakage currents as a function of electric field were collected by the same ferroelectric test system. In the thermally stimulated discharge current (TSDC, Novocontrol) test, a DC poling electric field of 15 MV/m was applied for 30 min. After poling, the samples were rapidly cooled down to -100 °C while holding the electric field. Afterwards the electric field was removed and the samples were short-circuited and heated to 100 °C at the heating rate of 5 °C/min while the current was measured across the sample.

4.5. Phase-field Simulations

A unipolar electron injection and transport model is established without consideration the effects of hole and recombination of bipolar charges, to qualitatively understand the behaviors of charge carriers in nanocomposites. Here, electrons are injected at the interface of dielectric/cathode

according to the Schottky mechanism. The apparent mobility μ_a is introduced to describe the trapping or detrapping process of mobile electrons with the function of $\mu_a = \mu_0 \exp\left(-\frac{\varphi_a}{k_B T}\right)$, where φ_a and μ_0 are the transport potential barrier and the initial electron mobility, and k_B and T are the Boltzmann constant and temperature, respectively. The potential barrier φ_a denotes how much energy need to be gained for a trapped electron becoming mobile. The higher φ_a is, the lower μ_a is. Here, the interfaces between two adjacent layers are regarded as a barrier layer hindering charge transport due to the depolarization field. Thus, the apparent mobility in the interfaces with a higher transport potential barrier is much lower than other areas.

The current density J_e at the cathode is described by

$$J_e = AT^2 \exp\left(-\frac{\varphi_i}{k_B T}\right) \exp\left(\frac{e}{k_B T} \sqrt{\frac{eE}{4\pi\epsilon}}\right)$$

where A the Richardson constant, T the temperature, φ_i the Schottky barrier for electron injection, e the elementary electron charge, E the electric field and ϵ the dielectric constant.

The current density $J(\mathbf{r}, t)$ is expressed by

$$J(\mathbf{r}, t) = \mu_a(\mathbf{r}, t) n(\mathbf{r}, t) E(\mathbf{r}, t)$$

where $n(\mathbf{r}, t)$ is the charge density of mobile electrons.

The behavior of charge carriers within the nanocomposites is then governed by Poisson's equation, and continuity equation, as follows:

$$\frac{\partial \psi(\mathbf{r}, t)}{\partial r_i \partial r_j} = -\frac{q(\mathbf{r}, t)}{\epsilon_0 \epsilon_r}$$

$$\frac{\partial J(\mathbf{r}, t)}{\partial \mathbf{r}} + \frac{\partial n(\mathbf{r}, t)}{\partial t} = s$$

where $\Psi(\mathbf{r}, t)$ the electrical potential, $q(\mathbf{r}, t)$ the free charge density (here equal $n(\mathbf{r}, t)$), s the source term ($s=0$ in this simulation and an apparent mobility is introduced to roughly reflect the trapping and detrapping process). **Table 2** presents the values of parameters used in this simulation at 400 K and 15 MV/m.

Table 2. The values of parameters used in this simulation at 400 K and 15 MV/m.

Parameters	Values in matrix	Values in filler
Schottky barrier (eV)	0.7	0.3
transport potential barrier ϕ_a (eV)	0.24 (Layer Interface)	≈ 0 (Other areas)
Electron mobility ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	9×10^{-11}	10
Dielectric constant	10	1000

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Figures

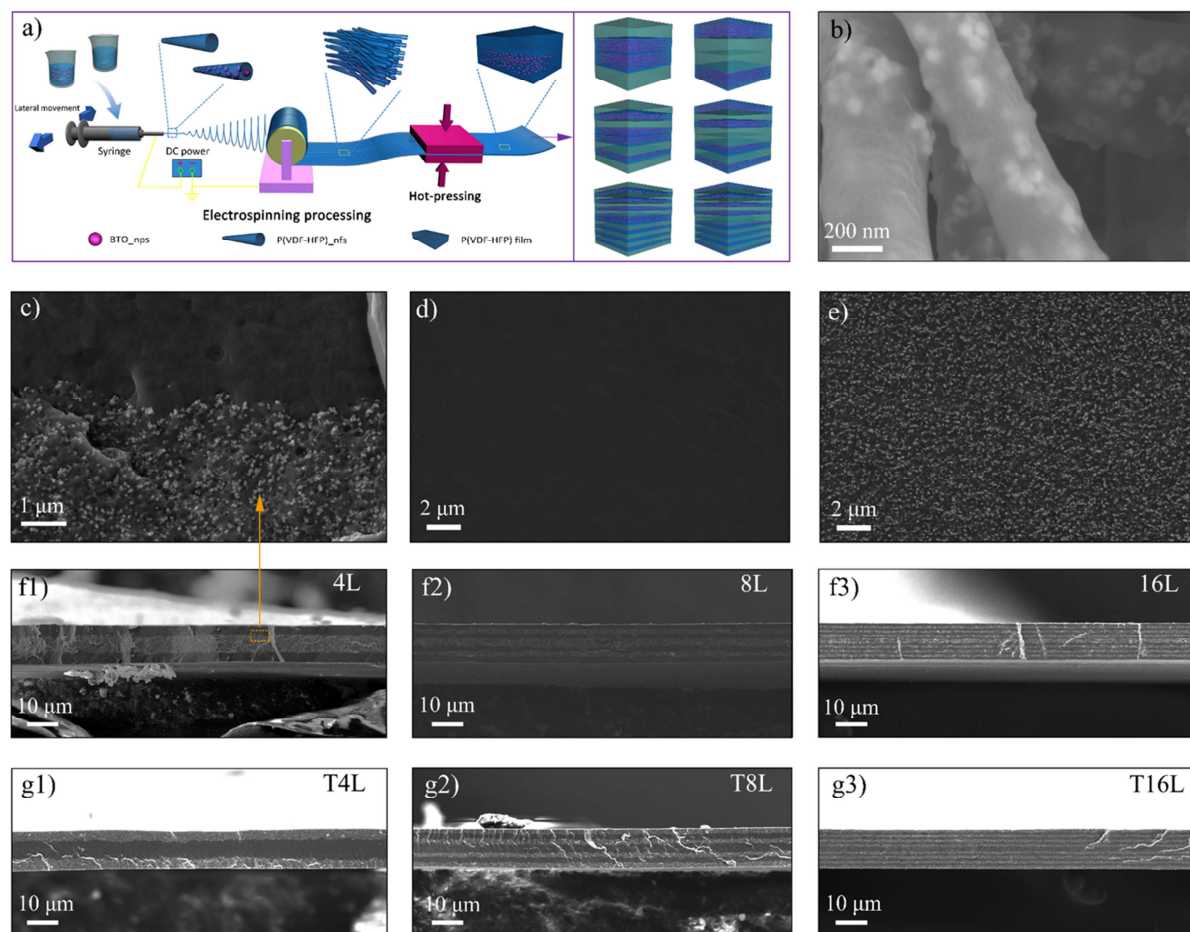


Figure 1. Fabrication and characterization multilayered nanocomposites. a) Schematic illustration of the synthesis procedure of the multilayered polymer nanocomposites using the non-equilibrium process (left) and six types of multilayered polymer nanocomposites (right). b) SEM image of P(VDF-HFP)-10 vol.% BTO nanofibers. c) Local magnified cross-sectional SEM image of the 4L film. Surface SEM images of d) the 16L and e) the T16L films. Cross-sectional SEM images of the f1) 4L, f2) 8L, f3) 16L, g1) T4L, g2) T8L and g3) T16L films, respectively.

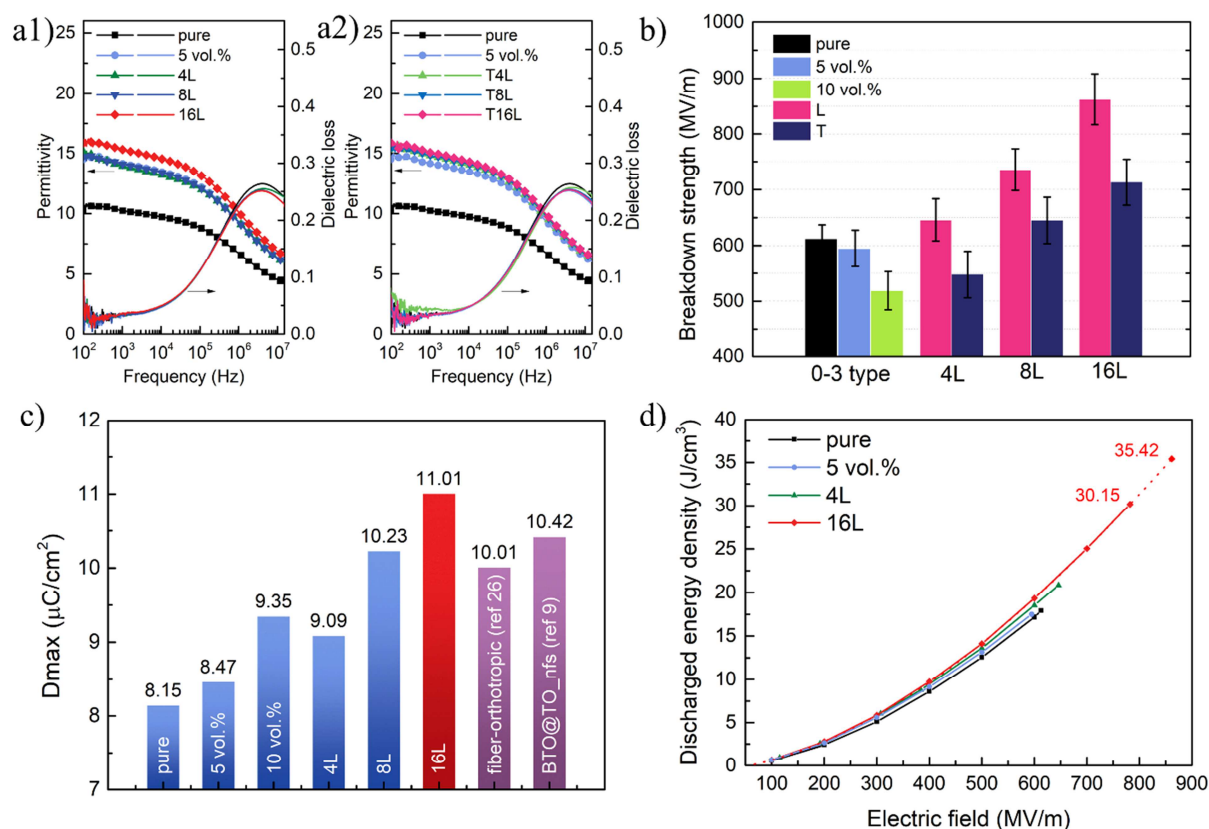


Figure 2. Dielectric performances of multilayered nanocomposites. a1, a2) Frequency dependence of permittivity of multilayered nanocomposites. b) Breakdown strength of multilayered nanocomposites and the controls. c) Comparison of the maximum electric displacement (at E_b) of multilayered nanocomposites and the controls in this work and two polymer nanocomposites reported in the previous work.

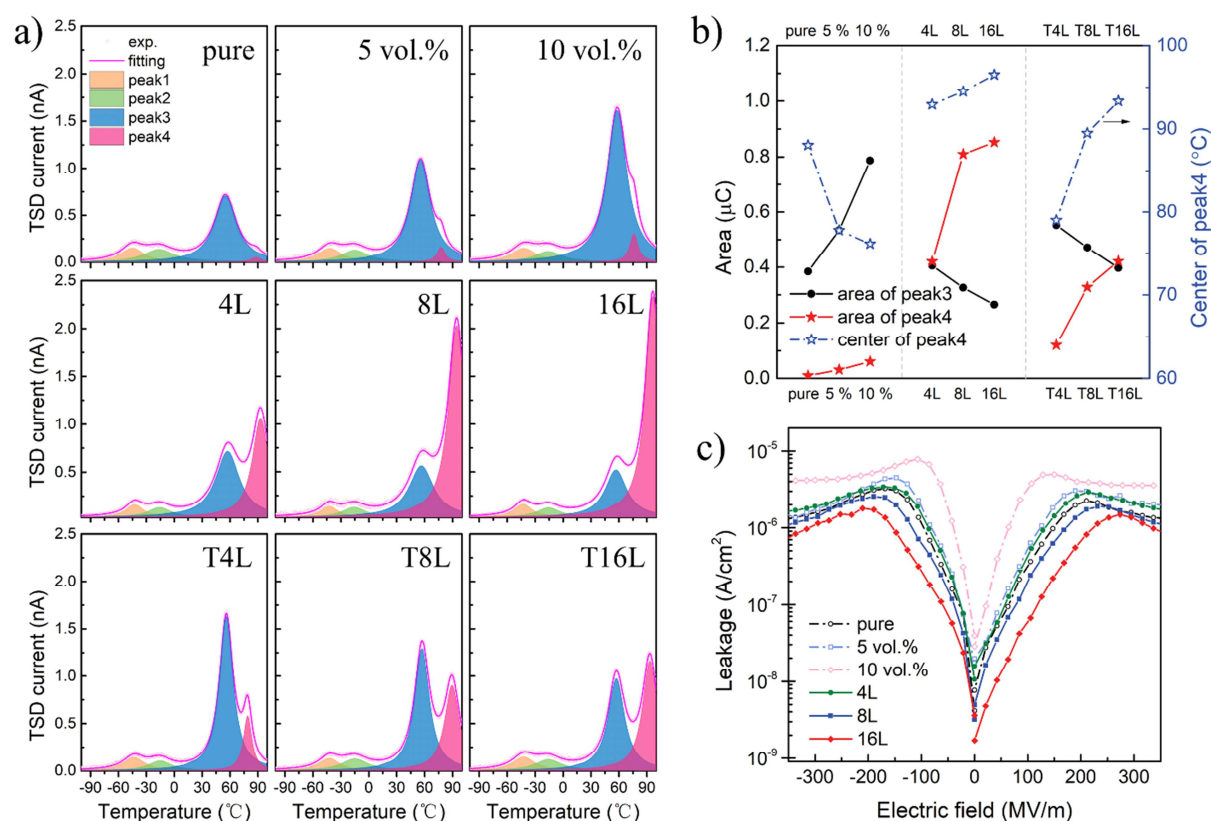


Figure 3. a) TSDC spectra for multilayered nanocomposites and the controls, and their corresponding peak-fitting results. b) Comparison of area and center of peak 3 and peak 4. c) Leakage current for multilayered nanocomposites and the controls as a function of electric field.

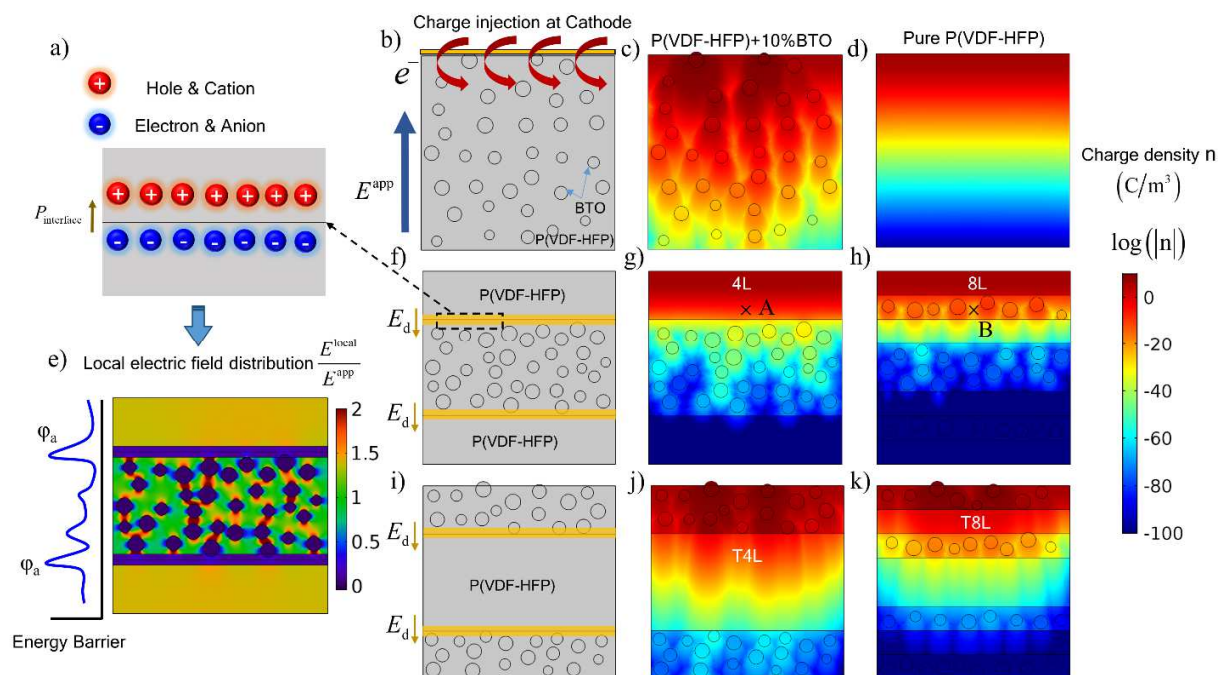
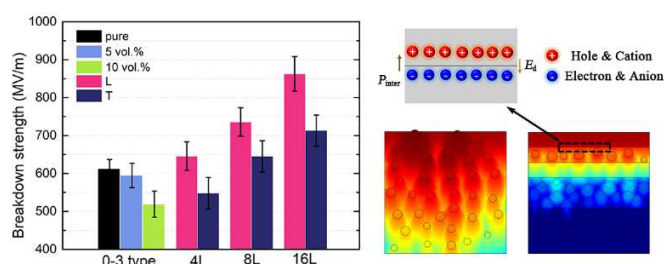


Figure 4. Phase-field simulation for multilayered nanocomposites. Schematic illustration of a) polarized interfacial ions and the induced depolarized electric field, b) 0-3 type P(VDF-HFP)-BTO nanocomposites, f) the 4L nanocomposites and i) the T4L nanocomposites, respectively. The distribution of charge density in c) 0-3 type P(VDF-HFP)-10 vol.% BTO, d) pure P(VDF-HFP), g) the 4L nanocomposites, h) the 8L nanocomposites, j) the T4L nanocomposites, k) the T8L nanocomposites, respectively. e) Local electric field distribution in the 4L nanocomposites.

Table 1. Comparison of the discharged energy density (U_e) and efficiency (η) at E_b in this work and the related polymer nanocomposites reported in the previous literatures.

Samples	U_e (J/cm ³)	η (%)	References
0-3 type BT_nps/P(VDF-HFP) nanocomposites	8.31	56.64	Ref. 28
BTO nanocrystals/P(VDF-HFP) nanocomposites	9.7	52	Ref. 12
BT@SiO ₂ _nps/PVDF nanocomposites	11.2	64	Ref. 31
BT@BN/PVDF nanocomposites	17.6	71.4	Ref. 32
TiO ₂ nanorod array/PVDF nanocomposites	10.62	69	Ref. 33
Sandwiched BT/PVDF nanocomposites	18.8	65	Ref. 17
Sandwiched NbNO ₃ platelets/PVDF	13.5	66.9	Ref. 2
Multilayered nanocomposites (16L)	30.15	78	This work

TOC



Graphical abstract

PVDF-based multilayered nanocomposites (up to 16 layers) are fabricated by the non-equilibrium process. Compared to the traditional 0-3 type nanocomposites, multilayered nanocomposites exhibit significantly enhanced breakdown strength (up to 862 MV/m), which is attributed to the suppression of the adverse effects of dielectric/electrode and dielectric/dielectric interfaces on the injection and migration of charges from the electrodes. This greatly increased breakdown strength gives rise to an unprecedentedly high discharge energy density (expected to be $\sim 35.4 \text{ J/cm}^3$).

Research highlights

- PVDF-based polymer nanocomposites with multilayered structure (up to 16 layers) are demonstrated and fabricated by the non-equilibrium process
- Synergy of micro-/mesoscopic interfaces in multilayered polymer nanocomposites leads to a greatly enhanced breakdown strength and an unprecedentedly high discharge energy density (expected to be $\sim 35.4 \text{ J/cm}^3$).
- TSDC and phase-field simulation are also employed to investigate the mechanisms about charge transportation in multilayered nanocomposites.



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- An Unprecedentedly high discharge energy density of $\sim 35 \text{ MJ/m}^3$ is achieved in a multilayered polymer nanocomposite.
- The rationally designed multilayered polymer nanocomposites are capable of integrating the suppressed effects of dielectric/electrode and dielectric/dielectric interfaces on charge injection and migration to remarkably enhance the breakdown strength
- A phase-field model reveals that the optimized topological structure and phase-composition induce favorable distribution of local electric field hence give rise to the ultrahigh discharge efficiency.